Applied AI & Machine Learning
CS-333

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PNEC, NUST

Lecture 6

Spring 2026

Intro to Computer Vision

Learning Outcomes

By the end of this lecture, students will be able to:

- Define Computer Vision and explain its role within Artificial Intelligence
- Describe the basic Computer Vision pipeline
- Explain core Computer Vision tasks
- Distinguish between traditional vision methods and deep learning-based approaches
- Identify real-world applications of Computer Vision in engineering and industry
- Recognize key challenges in Computer Vision systems

WHAT IS COMPUTER VISION?

HAVE YOU EVER USED IT?

Laptop: Biometrics auto-login (face recognition, 3D), OCR

Smartphones: QR codes, computational photography (Android Lens Blur, iPhone Portrait Mode), panorama construction (Google Photo Spheres), face detection, expression detection (smile), Snapchat filters (face tracking), Google Tango (3D reconstruction), Night Sight (Pixel)

Web: Image search, Google photos (face recognition, object recognition, scene recognition, geolocalization from vision), Facebook (image captioning), Google maps aerial imaging (image stitching), YouTube (content categorization)

VR/AR: Outside-in tracking (HTC VIVE), inside out tracking (simultaneous localization and mapping, HoloLens), object occlusion (dense depth estimation)

Motion: Kinect, full body tracking of skeleton, gesture recognition, virtual try-on

Medical imaging: CAT / MRI reconstruction, assisted diagnosis, automatic pathology, connectomics, endoscopic surgery

Industry: Vision-based robotics (marker-based), machine-assisted router (jig), automated post, ANPR (number plates), surveillance, drones, shopping

Transportation: Assisted driving (everything), face tracking/iris dilation for drunkeness, drowsiness, automated distribution (all modes)

Media: Visual effects for film, TV (reconstruction), virtual sports replay (reconstruction), semantics-based auto edits (reconstruction, recognition)

Source: https://cs.brown.edu/courses/csci1430/lectures/2019Spring_01_Introduction.pdf

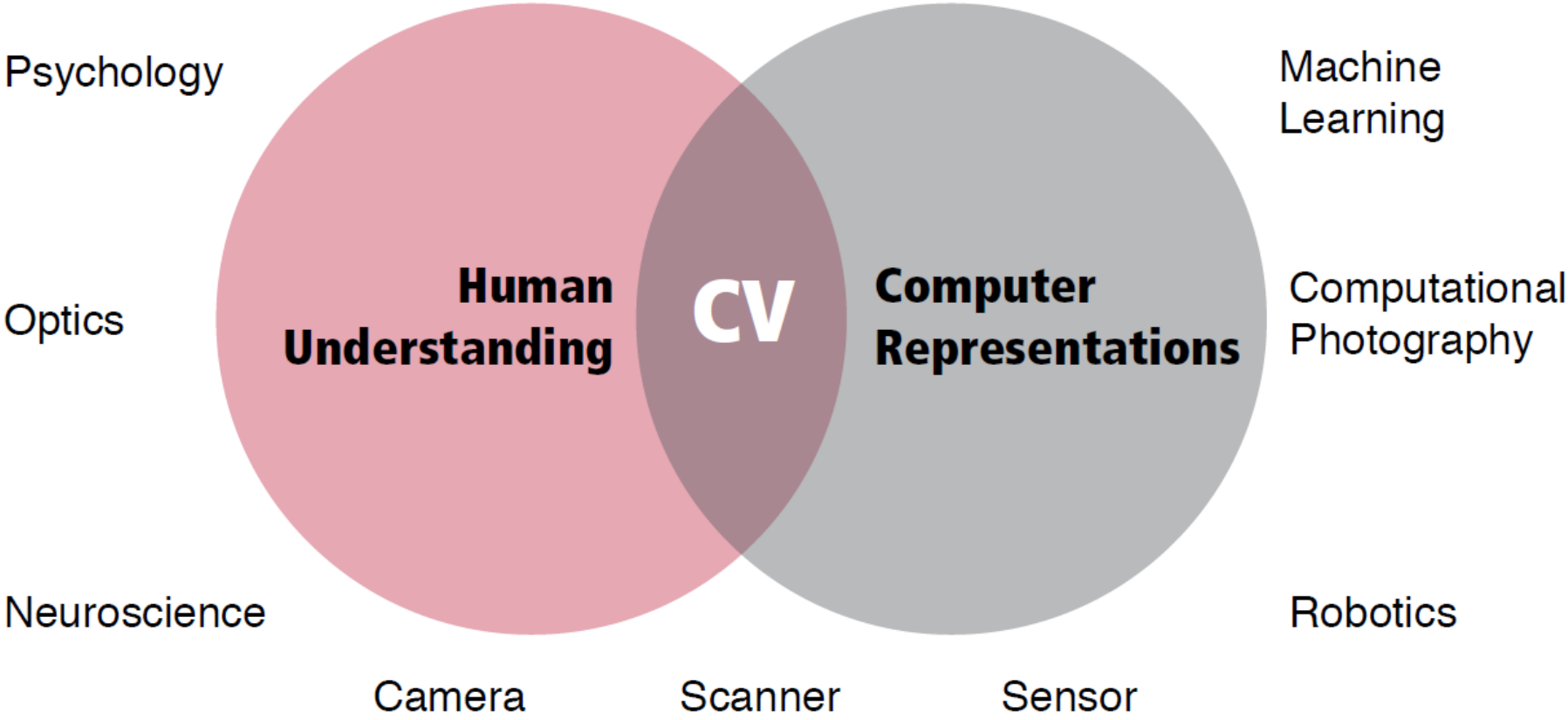
Most probably…

DEFINITION

"Computer vision is concerned with
the automatic extraction, analysis & understanding of useful information
from a single image or a sequence of images.
It involves the development of **a theoretical and algorithmic basis**
to achieve automatic visual understanding."

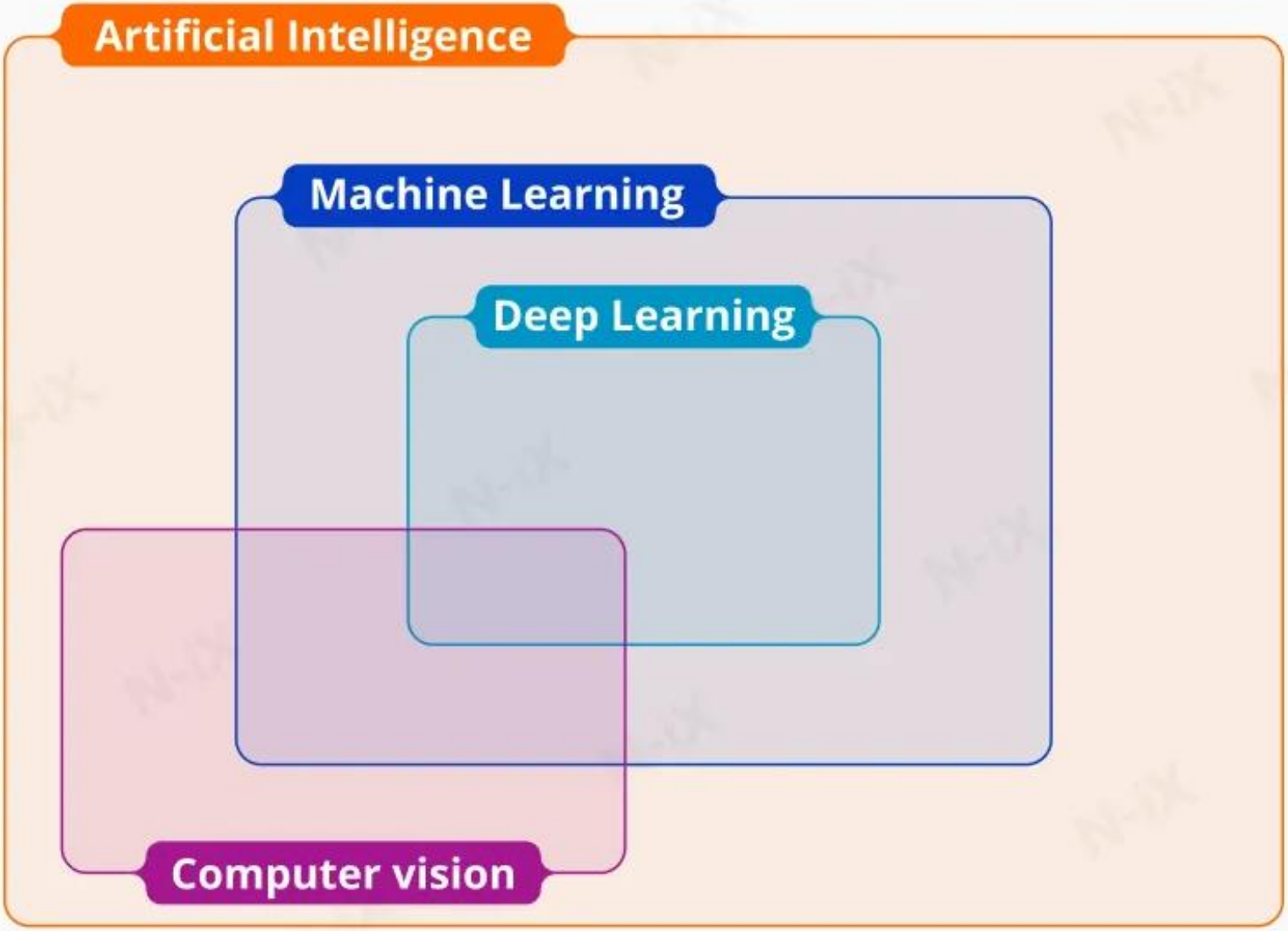
Attributed to The British Machine Vision Association and Society for Pattern Recognition

MULTIDISCIPLINARY



A brief history of computer vision

- 1960s - started as a student summer project at MIT.
- 1970s and 80s – part of AI – understanding human vision and emulating human perception.
- 1990s – depart from AI , geometric approach.
- Today – various mathematical methods (statistics, differential equations, optimization), applications (security, robotics, graphics).


Artificial Intelligence
Machine Learning
Deep Learning
Computer vision

Every image tells a story

- Goal of computer vision: perceive the "story" behind the picture
- Compute properties of the world
  - 3D shape
  - Names of people or objects
  - What happened?

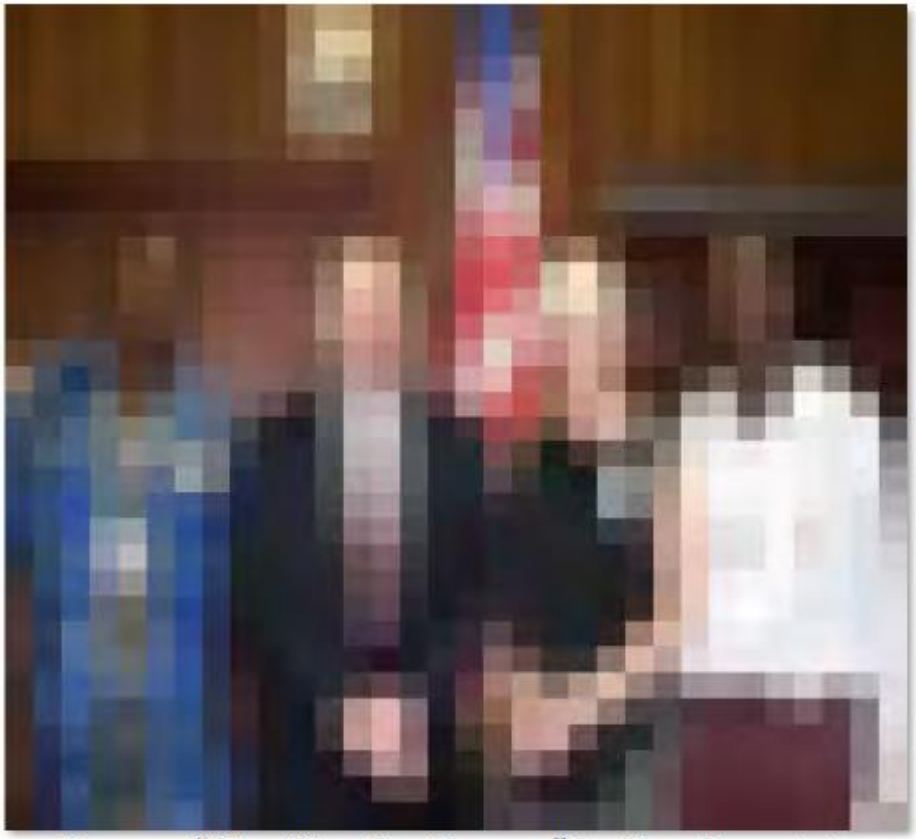
The goal of computer vision

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Can the computer match human perception?

- Yes and no (mainly no)
  - computers can be better at "easy" things
  - humans are much better at "hard" things
- But huge progress has been made
  - Especially in the last 10 years
  - What is considered "hard" keeps changing

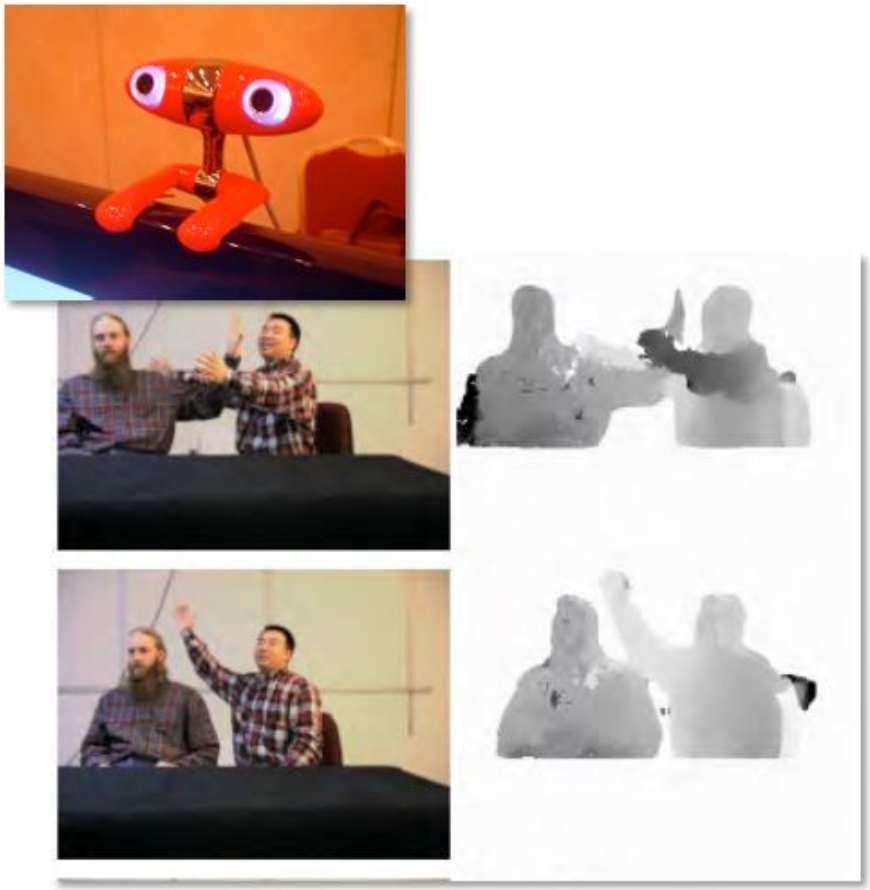
But humans can tell a lot about a scene from a little information…

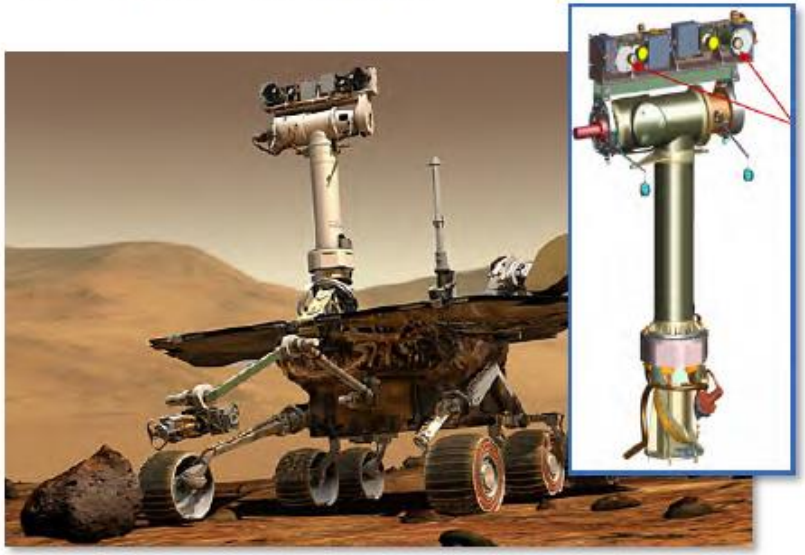


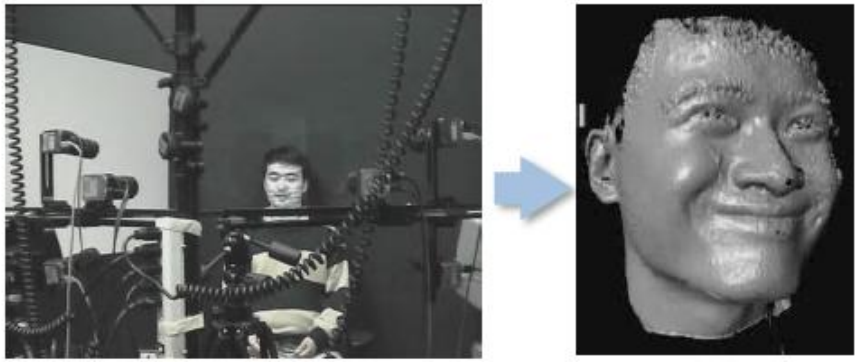
Source: "80 million tiny images" by Torralba, et al.

The goal of computer vision

- Computing the 3D shape of the world



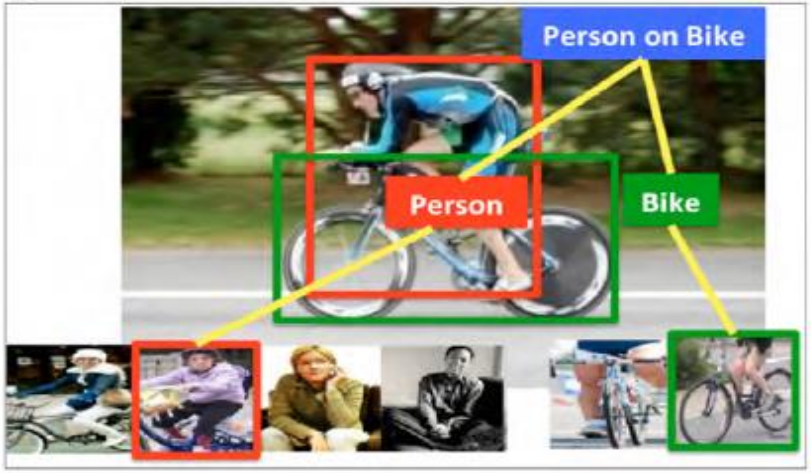




The goal of computer vision

- Recognizing objects and people

- Object detection
- Action classification
- Image captioning
- …

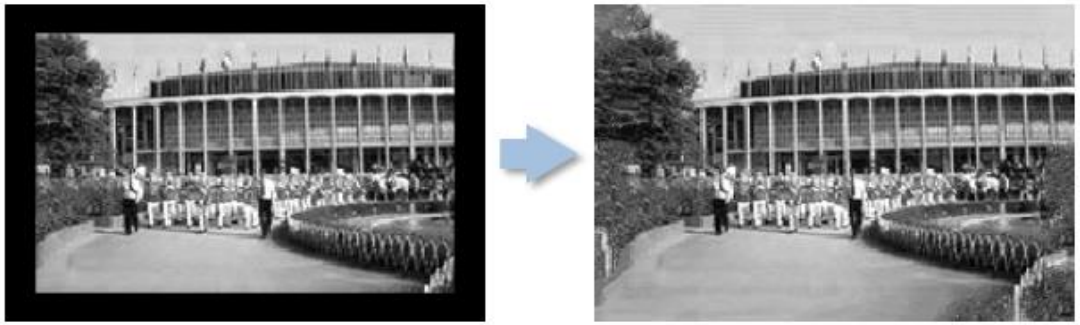


sky
building
flag
face
banner
wall
street lamp
bus
bus
cars
slide credit: Fei-Fei, Fergus & Torralba

The goal of computer vision

- "Enhancing" images (c.f. Computational Photography)

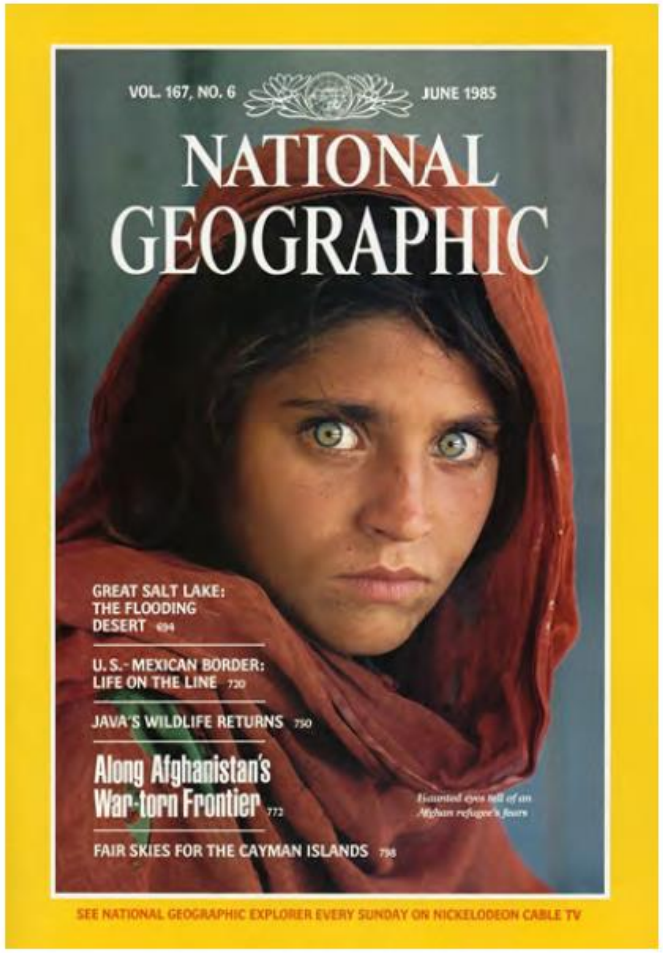
Super-resolution / denoising
(source: 2d3)



Texture synthesis / increased field of view (uncropping)
(image credit: Efros and Leung)

Inpainting / image completion
(image credit: Hays and Efros)

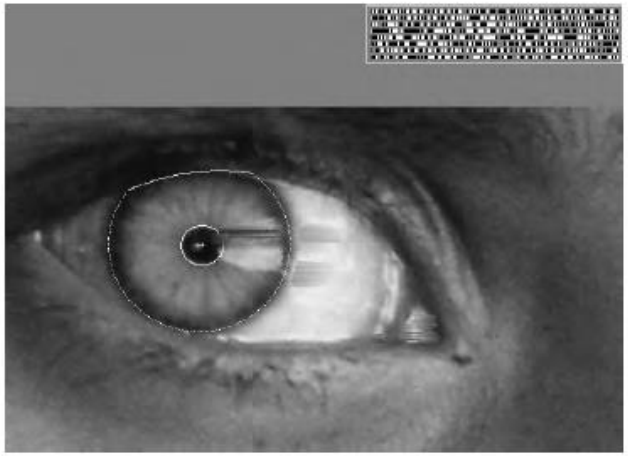
Face recognition

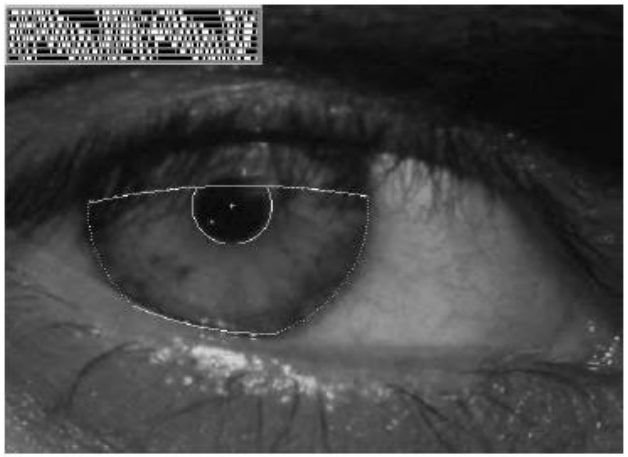


Who is she?

Vision-based biometrics

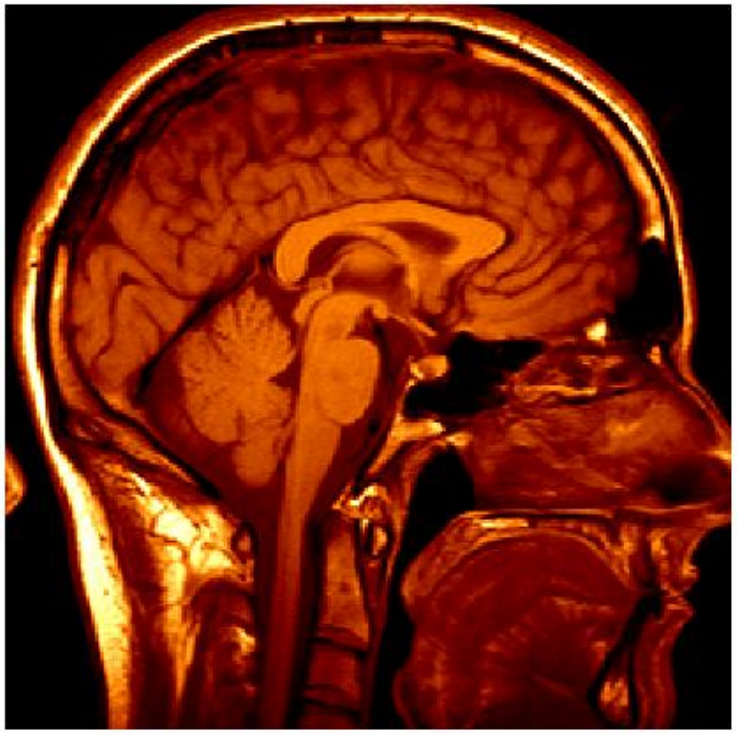
"*How the Afghan Girl was Identified by Her Iris Patterns*" Read the story





Source: S. Seitz

Medical imaging



3D imaging
MRI, CT

Image guided surgery
Grimson et al., MIT

Why is computer vision difficult?

Viewpoint variation

Illumination

Scale

Why is computer vision difficult?

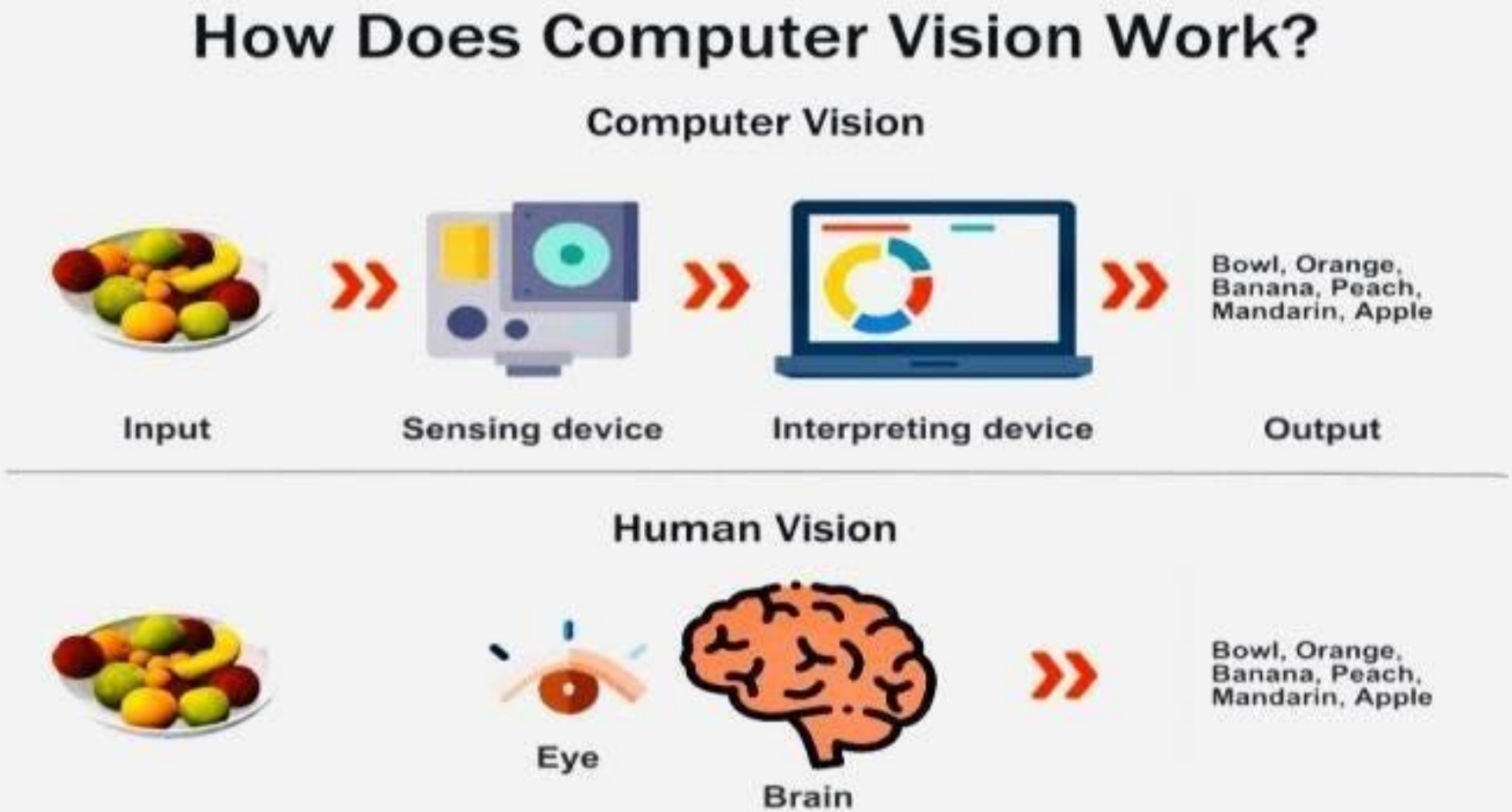
Intra-class variation

Motion (Source: S. Lazebnik)

Background clutter

Occlusion

HOW COMPUTER VISION WORK?


How Does Computer Vision Work?
Computer Vision
Bowl, Orange, Banana, Peach, Mandarin, Apple
Input
Sensing device
Interpreting device
Output
Human Vision
Bowl, Orange, Banana, Peach, Mandarin, Apple
Eye
Brain

Core Computer Vision Tasks

Low-Level Vision
- Image processing
- Feature detection

Mid-Level Vision
- Segmentation
- Motion analysis

High-Level Vision
- Object recognition
- 3D reconstruction

Low-Level Vision

Image (Pre-)Processing

Low-level operations.

Examples:

•Noise removal

•Filtering

•Contrast enhancement

Feature Detection

Extract important structures.

Examples:

•Edges

•Corners

•Keypoints